J. Dairy Sci. 106:5115–5126
https://doi.org/10.3168/jds.2022-22651
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Presynchronization with prostaglandin $F_{2\alpha}$ and gonadotropin-releasing hormone simultaneously improved first-service pregnancy per artificial insemination in lactating Holstein cows compared with Presynch-14 when combined with detection of estrus

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ABSTRACT

This study aimed to determine the effect of 2 simple breeding strategies combining artificial insemination (AI) after detection of estrus (AIED) and timed AI (TAI) on first-service fertility in lactating Holstein cows. Weekly, lactating Holstein cows (n = 1,049) between 40 and 46 d in milk (DIM) were randomly assigned to initiate 1 of 2 breeding strategies for first service: Presynch-14 and PG+G. Presynch-14 is a presynchronization strategy with 2 $PGF_{2\alpha}$ treatments 14 d apart with the last $PGF_{2\alpha}$ 14 d before the initiation of the Ovsynch protocol. Cows treated with PG+G receive a simpler presynchronization program that uses $PGF_{2\alpha}$ and GnRH simultaneously 7 d before Ovsynch. In both treatments, cows detected in standing estrus by tail chalk at any time ≥55 DIM were inseminated, and treatment was discontinued (n = 525). Cows completing treatment received TAI from 78 to 84 DIM (n = 526). In a subgroup of cows that received TAI, blood was collected (n = 163) to assess circulating concentrations of progesterone, and ultrasonographic evaluations of ovaries were performed on the day of first GnRH of Ovsynch (n = 162) and $PGF_{2\alpha}$ of Ovsynch (n = 122). The proportion of cows that received TAI was greater for PG+G compared with Presynch-14 (63.5 vs. 31.9%), which increased DIM at first service for cows treated with PG+G compared with Presynch-14 (75.5 ± 0.4 vs. 68.7 ± 0.4). For cows receiving TAI, the ovulatory response to first GnRH of Ovsynch (73.8 vs. 48.8%) and the proportion of cows with functional corpora lutea (92.6 vs. 73.1%) were greater for PG+G than Presynch-14. Cows treated with PG+G had greater overall pregnancy per AI (P/AI) 42 ± 7 d after AI (40.2 vs. 33.6%) and calving per AI (32.1 vs. 25.2%) than Presynch-14. For cows receiving AIED, treatment did not affect P/AI 42 ± 7 d after AI. However, for cows receiving TAI, PG+G increased P/AI compared with Presynch-14 (44.6 vs. 35.2%). Overall, cows receiving TAI had greater P/AI 42 ± 7 d after AI (42.5 vs. 31.5%) and calving per AI (34.1 vs. 23.7%) and decreased pregnancy loss (16.8 vs. 25.2%) than cows receiving AIED. In summary, PG+G increased the proportion of cows receiving TAI and the DIM at first service, P/AI, and calving per AI compared with Presynch-14 when both TAI programs were combined with AIED.

Key words: presynchronization, Ovsynch, detection of estrus, fertility, timed artificial insemination

INTRODUCTION

Fertility programs such as Presynch-10 or -11 (Galvão et al., 2007; Stevenson, 2011; Stevenson and Pulley, 2012), G6G (Bello et al., 2006), and Double-Ovsynch (Souza et al., 2008) improve AI submission rate and pregnancy per AI (**P/AI**) compared with insemination following estrus for the first service postpartum in high-producing lactating dairy cows (Gumen et al., 2012; Santos et al., 2017). These programs also allow precise control of the voluntary waiting period (**VWP**; Martins and Pursley, 2016; Carvalho et al., 2018). However, from a management perspective, implementing these programs in dairy operations can be logistically challenging due to hormone treatments and timed AI (**TAI**) on 5 different days of the week, increasing labor requirements and the risk of program compliance issues. To simplify reproductive breeding management, some commercial dairy producers in the United States prefer to use a presynchronization strategy with 2 $PGF_{2\alpha}$ treatments 14 d apart with the last $PGF_{2\alpha}$ 14 d before the initiation of the Ovsynch protocol (**Presynch-14**; Navanukraw et al., 2004). This protocol design concentrates most treatments on the same day of

Received August 14, 2022.
Accepted January 6, 2023.
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the week, except for the additional $PGF_{2\alpha}$ and the last GnRH of Ovsynch. Implementing a TAI protocol with most treatments on the same day can also coincide with the day of pregnancy diagnosis, offering opportunities for treatments to be completed by the veterinarian or having the veterinarian more directly involved with overseeing compliance of treatments.

Moreover, US dairy producers often combine this Presynch-Ovsynch strategy with AI after detection of estrus (**AIED**) following the second $PGF_{2\alpha}$ treatment of the presynchronization, which typically aligns with the end of the VWP. In herds with effective detection of estrus, this breeding management scheme allows the insemination of a high percentage of cows (~50 to ~70%) between the end of the VWP and the first GnRH of Ovsynch (**G_1**; Lima et al., 2009; Fricke et al., 2014; Giordano et al., 2016). However, this strategy might limit fertility outcomes. Cows inseminated following estrus may have fewer P/AI to the first service postpartum compared with cows only receiving TAI following a fertility program (Gumen et al., 2012; Fricke et al., 2014; Santos et al., 2017). In addition, a high proportion of cows that do not receive AIED do not ovulate at G_1 and are not synchronized for the TAI, reducing the probability of conceiving (Galvão et al., 2007; Galvão and Santos, 2010). Lastly, presynchronization strategies that only use $PGF_{2\alpha}$ do not induce cyclicity in anovular cows before Ovsynch (Moreira et al., 2001), which results in lower P/AI after TAI compared with presynchronization protocols that use GnRH (Herlihy et al., 2012).

Recently, a simpler presynchronization program that uses $PGF_{2\alpha}$ and GnRH simultaneously 7 d before Ovsynch (**PG+G**) was compared with 2 fertility treatments used for first service postpartum (Yousuf et al., 2016; Martins et al., 2017). The first study compared PG+G with G6G, which uses a GnRH treatment 2 d after $PGF_{2\alpha}$ and 6 d before Ovsynch (Bello et al., 2006; Yousuf et al., 2016). This presynchronization aims to increase the proportion of cows on d 6 of the estrous cycle at G_1, which is an optimal day to induce ovulation and control the onset of the ovulatory follicular wave (Vasconcelos et al., 1999; Martins and Pursley, 2016). Although G6G improved the proportion of cows presynchronized, ovulatory response to G_1 and P/AI at 35 and 49 d post-AI did not differ between cows treated with PG+G and G6G (Yousuf et al., 2016). In a second study, lactating dairy cows that received first-service TAI after PG+G also had similar P/AI at 28, 35, 49, and 77 d post-AI compared with cows that received TAI after Presynch-10 (Martins et al., 2017). Thus, PG+G may be a simple presynchronziation alternative that uses GnRH for dairy operations using Presynch-14. These dairy farms would not need to change the days of the week for hormone treatments and TAI, and herd reproductive performance may potentially improve.

Hence, this study objective was to compare reproductive physiological parameters during Ovsynch and first-service P/AI in lactating dairy cows treated with PG+G compared with Presynch-14 when both treatments were combined with AIED. In the present study, we also included an additional $PGF_{2\alpha}$ 21 d before the initiation of PG+G that would align with the same day of the week of the PG+G treatment. This additional $PGF_{2\alpha}$ treatment aimed to increase the proportion of cows at the end of the estrous cycle at the PG+G treatment and improve response to the $PGF_{2\alpha}$ and GnRH of the PG+G treatment. The study's primary hypothesis was that PG+G would increase ovulatory response to G_1 and serum progesterone (**P4**) concentrations at $PGF_{2\alpha}$ of Ovsynch and improve first-service P/AI compared with Presynch-14. Our second hypothesis was that PG+G would increase the proportion of cows receiving TAI compared with Presynch-14.

MATERIALS AND METHODS

The Institutional Animal Care and Use Committee at the University of California, Davis (Davis, CA) approved all procedures performed with cows in this experiment (Protocol #20091).

Cows, Farm, Management, and Products

This experiment was conducted on a commercial dairy farm in the California San Joaquin Valley (Fresno County), with approximately 2,800 lactating Holstein cows milked thrice daily. Lactating dairy cows (n = 1,119) were enrolled to receive treatments between July and December 2017. The average herd milk production during the experiment was 39 kg/cow per day. Milk yield was recorded at each milking using electronic flow meters (Metatron S21, Westfalia Surge Inc.). Cows were housed in 8 pens in 2 freestall barns with 4-row head-to-head stalls, concrete flooring, feedline soakers, and self-locking headlocks. Stalls were covered with deep dry-recycled solid-manure bedding. Each pen of the freestall had access to an outside dirt exercise lot. Cows were fed a TMR twice daily and had free access to feed and water. The TMR consisted of triticale silage, hay, almond hulls, steam-rolled corn, ground corn, citrus pulp, and a grain mix containing fuzzy cottonseed, distillers dried grains, molasses, corn gluten feed, canola, and mineral, vitamin, and protein supplements. The diets were formulated to meet or exceed nutrient recommendations for high-producing lactating Holstein cows (NRC, 2001).

Cows that were sold or died during treatment (n = 48), missed an injection of the treatment (n = 17), or were diagnosed with metritis by palpation at G_1 (n = 5) were removed from the study. A total of 1,049 cows were used in the study.

Cows were identified in estrus based on removal of tail chalk, which was applied to the tail head of all cows and evaluated once daily in the morning. Artificial insemination was conducted immediately after estrus detection. Estrus detection and AI were primarily performed by 2 AI technicians (n = 989) and occasionally by 2 relief AI technicians (n = 60). One AI technician was responsible for AI in 3 pens, and the other AI technician performed AI in the other 5 pens. All AI technicians were blinded to treatments. Commercial conventional frozen semen from multiple Holstein, Angus, or Limousin sires purchased by the farm was used for AI. Sires from beef breeds were used in 28.6% (300/1,049) of the total first-service inseminations in the study.

During hormone treatment days, the herd manager scanned electronic identification ear tags (Allflex) of all cows in the pens using an electronic identification reader and a handheld computer with Pocket CowCard software (DairyComp 305, Valley Ag Software) to identify cows that need $PGF_{2\alpha}$ or GnRH treatment. Intramuscular administration of $PGF_{2\alpha}$ (0.5 mg of cloprostenol; 2 mL of Estrumate, Merck Animal Health) and GnRH (86 μg of gonadorelin; 2 mL of Fertagyl, Merck Animal Health) were performed by herd veterinarian (J. S. Schmitt; Kings Veterinary Service, Leemore, CA), farm manager, trained farm employees, and trained personnel from our laboratory.

Treatments, Pregnancy Diagnoses, Culling, and Calving Records

Every week, lactating dairy cows from 40 to 46 DIM were automatically randomly assigned to 2 experimental groups (Figure 1) by the on-farm management software (DairyComp 305, Valley Ag Software). Cows assigned to the control group received Presynch-14 consisting of 2 $PGF_{2\alpha}$ treatments 14 d apart, with the second treatment 14 d before G_1 (Navanukraw et al., 2004). Cows assigned to PG+G received $PGF_{2\alpha}$ followed in 21 d with the administration of $PGF_{2\alpha}$ and GnRH concurrently 7 d before G_1. The concurrent treatments of $PGF_{2\alpha}$ and GnRH were administered in different injection sites a few seconds apart. The present study, different from previous studies using PG+G (Yousuf et al., 2016; Martins et al., 2017; Hubner et al., 2020), used a $PGF_{2\alpha}$ treatment 21 d before PG+G. This extra $PGF_{2\alpha}$ treatment was to potentially increase the proportion of cows with a responsive dominant follicle (**DF**) and corpus luteum (**CL**) at the time of PG+G. The Ovsynch-56 protocol consisted of GnRH at 71 ± 3 DIM, followed by $PGF_{2\alpha}$ 7 d later, then a second $PGF_{2\alpha}$ 1 d after the first $PGF_{2\alpha}$, the last GnRH 32 h later, and TAI approximately 16 h after the last GnRH (Pursley et al., 1995; Brusveen et al., 2008, 2009). Cows from both treatments detected in standing estrus by tail chalk at any time ≥55 DIM were inseminated and did not receive any additional treatment after AIED. Cows that completed Ovsynch-56 received TAI from 78 to 84 DIM. Cows detected in estrus 1 d before TAI (n = 20) were inseminated for 2 consecutive days.

Cows detected in estrus after AI were rebred and considered nonpregnant. A farm veterinarian blinded to treatments performed pregnancy diagnoses by transrectal palpation between 42 ± 7 d after AI and reconfirmed pregnancy status around 2 mo gestation. Cows detected in estrus or with abortion signs during gestation and reconfirmed nonpregnant status by transrectal palpation were considered to have undergone pregnancy loss on the day of the detection of estrus or abortion. Pregnancy loss, culling, and calving data were retrieved from individual cow cards of on-farm management software (DairyComp 305, Valley Ag Software).

Ovarian Ultrasonography

During wk 6 of the study, ovaries of cows enrolled in the Ovsynch-56 program were scanned by an experienced technician. Ultrasonographic evaluations of ovaries were performed on the day of G_1 (n = 166) and $PGF_{2\alpha}$ (n = 124) of Ovsynch using a portable ultrasound machine (Easi-Scan, BCF Technology) equipped with a 7.5-MHz linear-array transducer. The number and estimated average diameter of all follicles ≥9 mm and CL ≥15 mm were recorded on each ovary at G_1. On-screen background gridlines consisting of squares with 10 × 10-mm sides were used to estimate the average diameter size of follicles and CL. Ovulatory response to G_1 was determined by the presence of a large antral follicle (≥9 mm in diameter) at G_1 and the appearance of a new CL on the same ovary and location of the large follicle 7 d later at the $PGF_{2\alpha}$ of Ovsynch.

Blood Collection, Hormone Assays, and Analysis of Luteal Function

During wk 7 of the study, cows that initiated the Ovsynch-56 program had blood collected at the time of first GnRH (n = 165) and $PGF_{2\alpha}$ (n = 147) of Ovsynch by puncture of the coccygeal vein or artery using 10 mL Vacutainer tubes without anticoagulant (BD Vacutainer). Following collection, samples were placed in a cooler with ice and transported to the laboratory. Serum

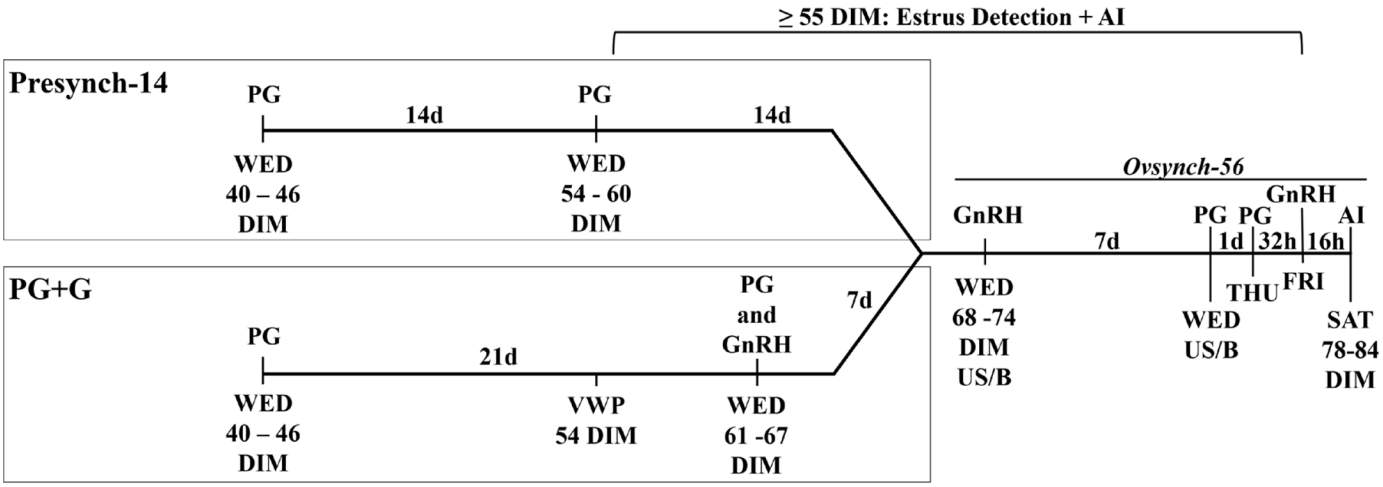


Figure 1. Schematic illustration of experimental procedures noting interval, DIM, and day of the week of hormonal treatments, $PGF_{2\alpha}$ (PG) and GnRH, blood collection (B), and ultrasonography examination of ovaries (US) for each treatment of Presynch-14 (control; n = 520) versus PG+G (n = 529). Lactating dairy cows were randomly assigned to initiate treatments at 40 to 46 DIM. Cows from both treatments detected in standing estrus by tail chalk any time after the voluntary waiting period (VWP = 54 DIM) were inseminated, and treatment was discontinued. Cows that completed treatment received timed AI from 78 to 84 DIM.

was separated within 6 h after collection by centrifugation at 2,000 × *g* for 20 min at room temperature and stored at −20°C. After completing on-farm data collection, all samples were placed in a cooler with dry ice and shipped for progesterone assay to the University of Wisconsin–Madison Department of Animal and Dairy Sciences laboratory. Serum P4 concentrations were quantified in a single assay using a solid-phase RIA kit containing antibody-coated tubes with 125I-labeled P4 (MP Biomedicals). The intra-assay coefficient of variation was 4.4%. Cows with serum concentrations of P4 ≥1.00 ng/mL were considered to have a functional CL.

Statistical Analyses

A priori statistical power analysis in G*Power version 3.1.9 (Faul et al., 2007) using the z-test to compute the sample size between 2 independent proportions, power = 0.80 and $\alpha = 0.05$, indicated that 932 animals were required to find a difference of 9 percentage points (36 vs. 45%) in P/AI between treatments in all types of insemination combined (TAI and AIED).

The experimental design used in this study was a completely randomized design. The effects of treatment (Presynch-14 vs. PG+G) and parity (primiparous vs. multiparous) on binomial variables (e.g., proportion of AIED or TAI, P/AI, pregnancy loss, and twinning rate) were analyzed by logistic regression with a generalized linear mixed model using the GLIMMIX procedure of SAS (version 9.4, SAS Institute, Inc.). In addition to treatment and parity, the initial model to analyze the effect of treatment on P/AI and calving per AI also considered the effects of the season (warm and cool), semen type (beef or Holstein), AI technician, and milk yield quartiles within parity groups. Variables in which $P > 0.20$ were removed from the model by backward stepwise elimination. Two-way interaction of treatment and parity was also considered in the final models if $P < 0.20$, and pairwise comparisons were performed using the LSmeans statement (SAS Institute). Our hypothesis was that PG+G would increase ovulatory response to G_1, the proportion of cows receiving TAI and first-service P/AI; therefore, a 1-tailed test was used to evaluate the effect of treatment on these parameters.

Differences between treatments in the proportion of cows inseminated on each day after the start of treatment were tested using Fisher's exact test using the FREQ procedure (SAS Institute). We used Fisher's exact test because the outcome was 0 or 100% in at least one of the treatment results from the variables listed.

The differences between treatments in continuous variables were analyzed using a linear mixed model applying the MIXED procedure of SAS with the same fixed effects (treatment and parity) as described for the GLIMMIX procedure. The assumptions for the normal distribution of the residuals and homoscedasticity of variance were tested with studentized residual plots for each variable. Residuals were considered normally distributed for all variables. A 1-tailed test was used to evaluate the effect of treatment on serum P4 concentrations at $PGF_{2\alpha}$ of Ovsynch.

For clarity, the actual means of the data were reported. Unless otherwise stated, analyses used a 2-tailed test. All values were considered different when $P \leq 0.05$, whereas $0.05 < P < 0.10$ were considered a statistical tendency.

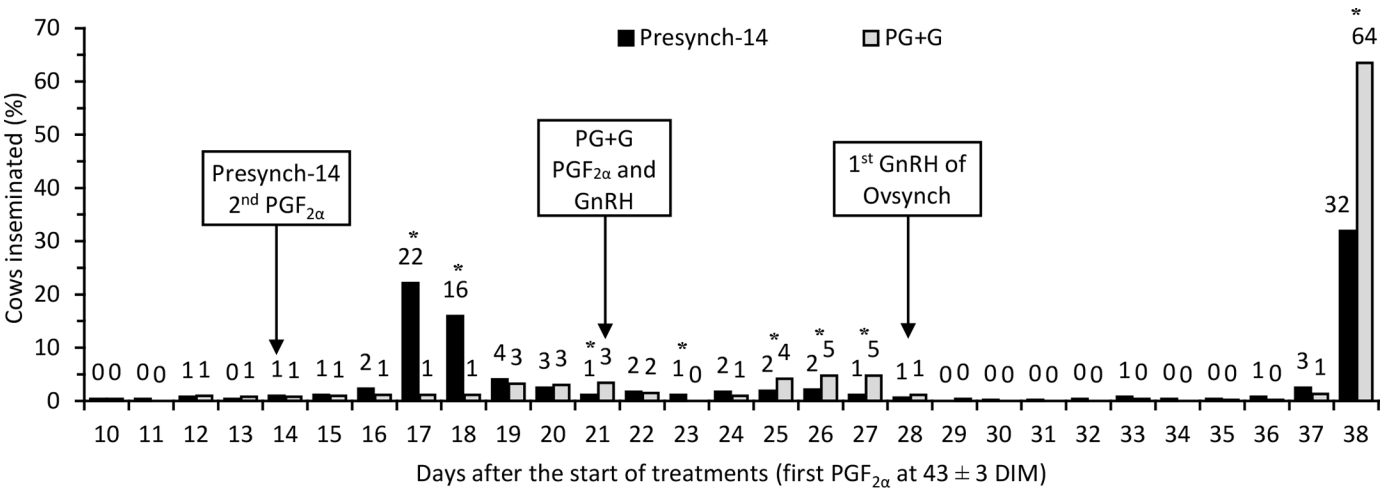


Figure 2. Distribution (% of total) of lactating dairy cows within treatments, Presynch-14 (control; n = 520) versus PG+G (n = 529), inseminated across days after the start of treatments (first $PGF_{2\alpha}$ at 43 ± 3 DIM). Cows treated with Presynch-14 received a second $PGF_{2\alpha}$ 14 d after the first $PGF_{2\alpha}$, and cows treated with PG+G received $PGF_{2\alpha}$ and GnRH treatment 21 d after the first $PGF_{2\alpha}$ injection. Cows from both treatments detected in standing estrus by tail chalk any time after ≥55 DIM were inseminated, and treatment was discontinued. Cows that completed treatment received timed AI (81 ± 3 DIM) 38 d after the start of treatments. Asterisks indicate differences ($P \leq 0.05$) among treatments in the proportion of cows inseminated on a specific day.

RESULTS AND DISCUSSION

Effect of Treatment on Insemination Pattern

The proportion of cows that received first-service AIED from 55 DIM to the day before TAI was 52.1% (547/1,049), while 47.9% (502/1,049) of the cows received only TAI for the first service. The majority of the cows that received AIED were inseminated from the end of VWP to the day of G_1 (92.3%; Figure 2). As expected, overall DIM at first AI for cows receiving AIED was earlier postpartum ($P < 0.001$) compared with cows receiving TAI (64.0 ± 0.2 vs. 81.0 ± 0.1 DIM).

A treatment by parity interaction for the proportion of cows receiving TAI was not observed ($P = 0.88$). Parity also did not affect ($P = 0.38$) the proportion of cows receiving TAI (47.0% primiparous vs. 48.4% multiparous). Nevertheless, PG+G increased the proportion of cows that received TAI compared with Presynch-14 (Figure 3), which confirmed our second hypothesis and resulted in a delayed ($P < 0.01$) DIM at first service postpartum for cows treated with PG+G compared with Presynch-14 (75.5 ± 0.4 vs. 68.7 ± 0.4 DIM, respectively). The decrease in the proportion of cows receiving AIED in PG+G could be explained by the design of the PG+G treatment. Most cows that responded to the first $PGF_{2\alpha}$ of the presynchronization (43 ± 3 DIM) were presumably in estrus 3 to 5 d later, before the end of the VWP (54 DIM), and most likely did not return to estrus before the $PGF_{2\alpha}$ and GnRH simultaneous treatments. Cows that did not respond to this first $PGF_{2\alpha}$ could be anovular or have an immature CL refractory to $PGF_{2\alpha}$ and likely did not show estrus before $PGF_{2\alpha}$ and GnRH. In addition, cows that responded to both $PGF_{2\alpha}$ and GnRH would most likely not express estrus before TAI.

Figure 2 illustrates the distribution of AI relative to the day after the start of treatments. A greater proportion of cows in Presynch-14 were inseminated on d 17 and 18 after the start of treatment compared with PG+G, which corresponded to d 3 and 4 after the sec-

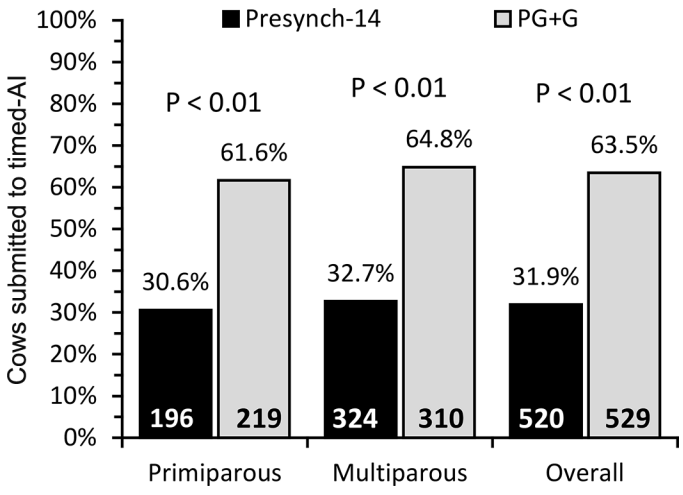


Figure 3. Effect of treatment, Presynch-14 (control) versus PG+G, on the proportion of cows submitted only to timed AI. Presynch-14 = a presynchronization strategy with 2 $PGF_{2\alpha}$ treatments 14 d apart with the last $PGF_{2\alpha}$ 14 d before the initiation of the Ovsynch protocol; PG+G = a simpler presynchronization program that uses $PGF_{2\alpha}$ and GnRH simultaneously 7 d before Ovsynch.

ond $PGF_{2\alpha}$ treatment of Presynch-14. Most likely, the cows detected in estrus on these days had a mature CL and a DF on the day of the second $PGF_{2\alpha}$ and responded to treatment. Previous studies using 2 $PGF_{2\alpha}$ treatments 14 d apart reported a similar spike in AIED on d 3 and 4 after the second $PGF_{2\alpha}$ (Pursley et al., 2012; Giordano et al., 2016). An increase in the proportion of cows treated with PG+G was mainly detected in estrus on d 25 to 27 after the start of the presynchronization, which corresponded to d 4 to 6 after $PGF_{2\alpha}$ and GnRH simultaneous treatments (Figure 2). Yousuf et al. (2016) reported that 14.2% (17/120) of cows treated with PG+G ovulated between d 2 and 7 after $PGF_{2\alpha}$ and GnRH treatments. In the present study, 14.6% of the cows treated with PG+G received AIED from d 3 to 6 after $PGF_{2\alpha}$ and GnRH treatments. These cows most likely had a combination of DF and CL absence or DF absence with a responsive CL in their ovaries on the day of $PGF_{2\alpha}$ and GnRH treatments (Yousuf et al., 2016).

A greater ($P = 0.01$) proportion of cows in Presynch-14 received AIED from the day after G_1 and the day before TAI compared with PG+G (5.6 vs. 2.5%, respectively). Furthermore, Presynch-14 tended to increase ($P = 0.06$) the proportion of cows receiving AIED from the day of the first $PGF_{2\alpha}$ of Ovsynch to the day before TAI compared with PG+G (3.7 vs. 1.7%, respectively).

For cows only receiving AIED, multiparous cows treated with PG+G (66.8 ± 0.5 DIM) had greater ($P < 0.04$) DIM at first AI postpartum compared with primiparous cows treated with PG+G (64.4 ± 0.6 DIM) and primiparous (63.4 ± 0.5 DIM) and multiparous (62.9 ± 0.5 DIM) cows treated with Presynch-14. Multiparous cows in Presynch-14 receiving AIED also had an earlier ($P = 0.03$) DIM mean at first AI postpartum compared with primiparous cows in PG+G.

Distribution of Beef Semen Among Treatments

The proportion of the total AI using sires from beef breeds did not differ between treatments ($P = 0.12$; 26.4% for Presynch-14 vs. 30.8% for PG+G) and parity ($P = 0.21$; 30.8% of primiparous vs. 27.1% of multiparous).

Milk Yield at AI

Daily average milk yield per cow for the week of AI did not differ ($P = 0.97$) for multiparous cows treated with Presynch-14 compared with PG+G (55.7 ± 0.4 vs. 55.8 ± 0.4 kg/d per cow, respectively). However, Presynch-14 primiparous cows tended ($P = 0.06$) to have higher milk yield than PG+G primiparous cows (42.4 ± 0.3 vs. 41.2 ± 0.3 kg/d per cow, respectively).

Effect of Treatment on Ovarian Parameters and Ovulatory Response in Cows Receiving TAI

Treatment did not affect the proportion of cows with a CL ≥15 mm in diameter or a functional CL (serum P4 ≥1.00 ng/mL) at G_1 (Table 1). A total of 74.7% of the cows had a CL with a diameter ≥15 mm at G_1, but only 58.8% had a functional CL (P4 ≥1.00 ng/mL). The high proportion of cows with CL at G_1 was anticipated since previous studies have demonstrated that 55 to 75% of cows not observed in estrus during presynchronization have a CL at G_1 (Fricke et al., 2014; Giordano et al., 2016). These cows may not have been inseminated because of an absence or lower intensity expression of estrus, a failure to detect cows in estrus after the end of the VWP, or both. The discrepancy between the 2 characterizations of luteal function, the proportion of cows with a visual CL during ultrasound examination of the ovaries and the proportion of cows with serum P4 ≥1.00 ng/mL, was consistent with previous studies (Silva et al., 2007; Stevenson et al., 2008; Giordano et al., 2016). In the present study, this discrepancy seemed more apparent in cows in PG+G than Presynch-14 (Table 1), likely caused by an increase in the proportion of cows with a newly formed CL not synthesizing enough P4 to achieve circulating P4 concentrations ≥1.00 ng/mL. This is further evidenced by the reduced serum P4 concentrations for PG+G cows at G_1 compared with Presynch-14 in cows with functional CL (2.39 ± 0.24 vs. 4.50 ± 0.37 ng/mL, respectively; Table 1). Yousuf et al. (2016) identified 75.6% of the cows treated with PG+G with a functional CL at G_1, which was defined by ovulation to the presynchrony GnRH or between 2 d after PG+G and G_1, and serum P4 concentrations ≥0.50 ng/mL at G_1. This result is almost identical to the proportion of cows with CL detected by ultrasound in PG+G in the present study (76.6%; Table 1).

In contrast, cows treated with Presynch-14 were most likely in a later stage of the estrous cycle (e.g., mid-diestrus) with a mature CL producing more P4 at G_1 than PG+G cows. The decrease in circulating P4 concentrations from G_1 to $PGF_{2\alpha}$ of Ovsynch (4.50 ± 0.37 at G_1 vs. 4.00 ± 0.47 ng/mL at $PGF_{2\alpha}$) and the low proportion of cows with functional CL at $PGF_{2\alpha}$ of Ovsynch in Presynch-14 cows (73.1%; Table 2) also indicated that these cows were in mid-diestrus and some may have had the onset of luteolysis before $PGF_{2\alpha}$ of Ovsynch. Our results for the characterization of luteal function at G_1 after Presynch-14 with AIED match the

Table 1. Effect of treatment (Presynch-14 vs. PG+G)[1] on luteal function and ovarian physiological parameters at the first GnRH of Ovsynch (G_1) and ovulatory response to G_1 in a subset of lactating Holstein cows receiving the first-service timed AI

Item	Presynch-14	PG+G	*P*-value
Cows with a CL with diameter ≥15 mm, % (n/n)	70.9 (39/55)	76.6 (85/111)	0.47
Cows with serum P4 ≥1.00 ng/mL, % (n/n)	66.1 (37/56)	55.0 (60/109)	0.16
Serum P4,[2] ng/mL ± SEM	4.50 ± 0.37	2.39 ± 0.24	<0.001
Cows with follicles ≥10 mm and <25 mm,[3] % (n/n)	96.4 (53/55)	97.3 (108/111)	0.76
Mean number of follicles ≥10 mm and <25 mm per cow ± SEM	2.1 ± 0.1	2.3 ± 0.1	0.45
Mean diameter of the largest follicle,[3] mm ± SEM	16.7 ± 0.4	15.9 ± 0.3	0.13
Cows with cyst ≥25 mm, % (n/n)	18.2 (10/55)	9.9 (11/111)	0.34
Ovulation to G_1, % (n/n)	48.8 (20/41)	73.8 (62/84)	<0.01
Double ovulation to G_1, % (n/n)	30.0 (6/20)	37.1 (23/62)	0.57
Mean pre-ovulatory follicle diameter in cows with single ovulation, mm ± SEM	16.9 ± 0.8	15.6 ± 0.5	0.18
Mean of the larger pre-ovulatory follicle diameter in cows with double ovulation, mm ± SEM	15.5 ± 2.6	16.9 ± 0.9	0.52
Mean of the smaller pre-ovulatory follicle diameter in cows with double ovulation, mm ± SEM	13.2 ± 2.6	13.7 ± 0.5	0.59

[1]Presynch-14 = a presynchronization strategy with 2 $PGF_{2\alpha}$ treatments 14 d apart with the last $PGF_{2\alpha}$ 14 d before the initiation of the Ovsynch protocol; PG+G = a simpler presynchronization program that uses $PGF_{2\alpha}$ and GnRH simultaneously 7 d before Ovsynch; P4 = progesterone; CL = corpus luteum.

[2]Only cows with serum P4 ≥1.00 ng/mL were used in this analysis.

[3]Only follicles ≥10 mm and <25 mm were used in this analysis.

results from Giordano et al. (2016), which reported 73.3% of cows with CL detected by ultrasound and 75.3% of cows with functional CL by serum P4 ≥1.00 ng/mL at G_1. The proportion of CL detected by ultrasound did not differ ($P = 0.13$) among primiparous and multiparous cows (84.4 vs. 72.6%, respectively). However, the proportion of cows with functional CL tended to be greater ($P = 0.06$) for primiparous than for multiparous cows (70.2 vs. 54.2%, respectively).

Cows treated with PG+G had a greater ovulatory response to G_1 than Presynch-14 (Table 1). This result was already expected since presynchronization strategies using only $PGF_{2\alpha}$, and with the last $PGF_{2\alpha}$ treatment 14 d before Ovsynch has been demonstrated to reduce the proportion of cows responding to G_1. The ovulatory response to G_1 in Presynch-14 was similar to that previously reported in other studies (46.4% in Giordano et al., 2016; 44.7% in Galvão et al., 2007). Cows that responded to the second $PGF_{2\alpha}$ treatment of the presynchronization but did not express estrus or were not detected in estrus would most likely be on d 0 of the estrous cycle 3 to 5 d after the $PGF_{2\alpha}$ treatment. These cows would be between d 9 to 11 of the estrous cycle at G_1, a period of the estrous cycle that produces a poor ovulatory response to GnRH due to the lack of a DF responsive to an LH surge (Vasconcelos et al., 1999). In addition, the high serum P4 concentrations at G_1 in Presynch-14 cows may have attenuated the magnitude of the GnRH-induced LH surge (Giordano et al., 2012). Moreover, $PGF_{2\alpha}$ does not induce cyclicity or control the onset of a follicular wave in anovular cows (Moreira et al., 2001). However, cows ovulating to

Table 2. Effect of treatment (Presynch-14 vs. PG+G)[1] on luteal function and ovarian physiological parameters at the $PGF_{2\alpha}$ of Ovsynch in a subset of first-service lactating Holstein cows

Item	Presynch-14	PG+G	*P*-value
Cows with a CL with diameter ≥15 mm, % (n/n)	87.8 (36/41)	98.8 (82/83)	0.98
Mean number of CL[2] ± SEM	1.6 ± 0.1	2.2 ± 0.1	<0.001
Cows with serum P4 ≥1.00 ng/mL, % (n/n)	73.1 (38/52)	92.6 (88/95)	<0.01
Serum P4,[3] ng/mL ± SEM	4.00 ± 0.47	5.12 ± 0.26	<0.01
Cows with follicles ≥10 mm and <25 mm, % (n/n)	97.6 (40/41)	98.8 (82/83)	0.61
Mean diameter of largest follicle,[4] mm ± SEM	15.2 ± 0.4	15.2 ± 0.3	0.37
Cows with cyst ≥25 mm, % (n/n)	26.8 (11/41)	10.8 (9/83)	0.03

[1]Presynch-14 = a presynchronization strategy with 2 $PGF_{2\alpha}$ treatments 14 d apart with the last $PGF_{2\alpha}$ 14 d before the initiation of the Ovsynch protocol; PG+G = a simpler presynchronization program that uses $PGF_{2\alpha}$ and GnRH simultaneously 7 d before Ovsynch; P4 = progesterone; CL = corpus luteum.

[2]Only cows with one or more CL ≥15 mm were used in this analysis.

[3]Only cows with serum P4 ≥1.00 ng/mL were used in this analysis.

[4]Only follicles ≥10 mm and <25 mm were used in this analysis.

the GnRH of the presynchronization of PG+G would be on approximately d 6 or 7 of the estrous cycle at G_1, which are days with high ovulatory response to a GnRH treatment (Vasconcelos et al., 1999; Bello et al., 2006). Previous studies reported ovulatory response to the GnRH of PG+G presynchronization from 58 to 70% and ovulation to G_1 from 68 to 91% (Yousuf et al., 2016; Hubner et al., 2020).

Although the ovulatory response to G_1 was increased in PG+G cows compared with Presynch-14, the proportion of cows with follicles from 10 to 25 mm in diameter was similar between treatments and overall high (97%). As previously indicated by Giordano et al. (2016), the simple presence of a large follicle >10 mm in diameter at G_1 is not an accurate predictor of ovulatory response to G_1. Treatment did not affect any other ovarian parameter related to follicle size at G_1 (Table 1). Furthermore, the proportion of cows with double ovulations did not differ among treatments, but it was high overall (35.4%). We already expected a high proportion of cows with double ovulations to G_1 in PG+G because this treatment increases the chance of cows growing ovulatory follicles in low circulating P4 concentrations (first follicular wave follicles), which induces codominance and double ovulation in high-producing lactating Holstein cows (Martins et al., 2018; Carvalho et al., 2019). The high proportion of cows with double ovulations to G_1 in Presynch-14 appears to be also caused by low serum P4 concentrations during the growth of the ovulatory follicle, which included anovular cows and cows during the dominance phase of the first follicular wave at G_1.

The proportion of cows with a CL detected by ultrasound at $PGF_{2\alpha}$ of Ovsynch did not differ between treatments; however, the PG+G treatment generated a greater proportion of cows with functional CL at $PGF_{2\alpha}$ of Ovsynch compared with Presynch-14 (Table 2). Potential reasons for this reduction in Presynch-14 are the increase in the proportion of cows with an immature CL recently formed or a regressing CL due to a decrease in the proportion of cows synchronized during Ovsynch. This is further supported by the lower circulating P4 concentration at $PGF_{2\alpha}$ of Ovsynch (Table 2), the discrepancy between both luteal parameters, and a greater proportion of cows receiving AIED during Ovsynch for Presynch-14 cows. The reduced ovulatory response to G_1 for Presynch-14 may have contributed to these results. Interestingly, the proportions of cows with CL by ultrasound (87.8%) and cows with functional CL (P4 ≥1.00 ng/mL; 73.1%) at $PGF_{2\alpha}$ of Ovsynch for Presynch-14 (Table 2) were nearly the same as reported by Giordano et al. (2016; 87.2 and 73.8%, respectively). The mean number of CL and serum P4 concentrations were higher for PG+G cows than for Presynch-14 (Table 2), likely due to the greater ovulatory response to G_1 in PG+G cows. The proportion of cows with cysts was lower for PG+G compared with Presynch-14, which may be due to the use of GnRH for the presynchronization in PG+G.

First-Service P/AI at 42 ± 7 d after AI, Pregnancy Losses, Calving, and Twinning

Primiparous cows had higher ($P < 0.001$) overall P/AI than multiparous cows (45.9 vs. 30.9%, respectively). This effect was similar ($P < 0.01$) in cows only receiving AIED or TAI. Primiparous cows appear to have higher fertility than multiparous cows. Other studies using only TAI (Martins et al., 2011; Herlihy et al., 2012), AIED (Santos et al., 2017), or a combination of both methods (Giordano et al., 2016) reported a similar parity effect on P/AI. Treatment also affected ($P = 0.04$) overall P/AI (Table 3). Cows treated with PG+G had greater overall P/AI than Presynch-14 (Figure 4 and Table 3).

Table 3. Effect of treatment (Presynch-14 vs. PG+G)[1] on pregnancy per AI (P/AI), calving per AI, pregnancy loss, and twin births

	Treatment		*P*-value		
Item	Presynch-14	PG+G	Treatment	Parity	Treatment × parity
P/AI at 42 ± 7 d after AI, % (n/n)	33.6 (171/509)	40.2 (206/512)	0.02	<0.001	0.86
Calving per AI,[2] % (n/n)	25.2 (126/500)	32.1 (163/508)	0.01	<0.001	0.83
Pregnancy loss,[2] % (n/n)	22.2 (36/162)	19.3 (39/202)	0.24	0.16	0.57
Twin births, % (n/n)	16.7 (21/126)	14.1 (23/163)	0.55	0.57	0.36

[1]Presynch-14 = a presynchronization strategy with 2 $PGF_{2\alpha}$ treatments 14 d apart with the last $PGF_{2\alpha}$ 14 d before the initiation of the Ovsynch protocol; PG+G = a simpler presynchronization program that uses $PGF_{2\alpha}$ and GnRH simultaneously 7 d before Ovsynch.

[2]Pregnancy losses from pregnancy diagnosis at 42 ± 7 d after AI to 245 d of gestation. Cows that left the herd after pregnancy diagnosis (n = 13) were not included in the analyses for calving per AI and pregnancy loss.

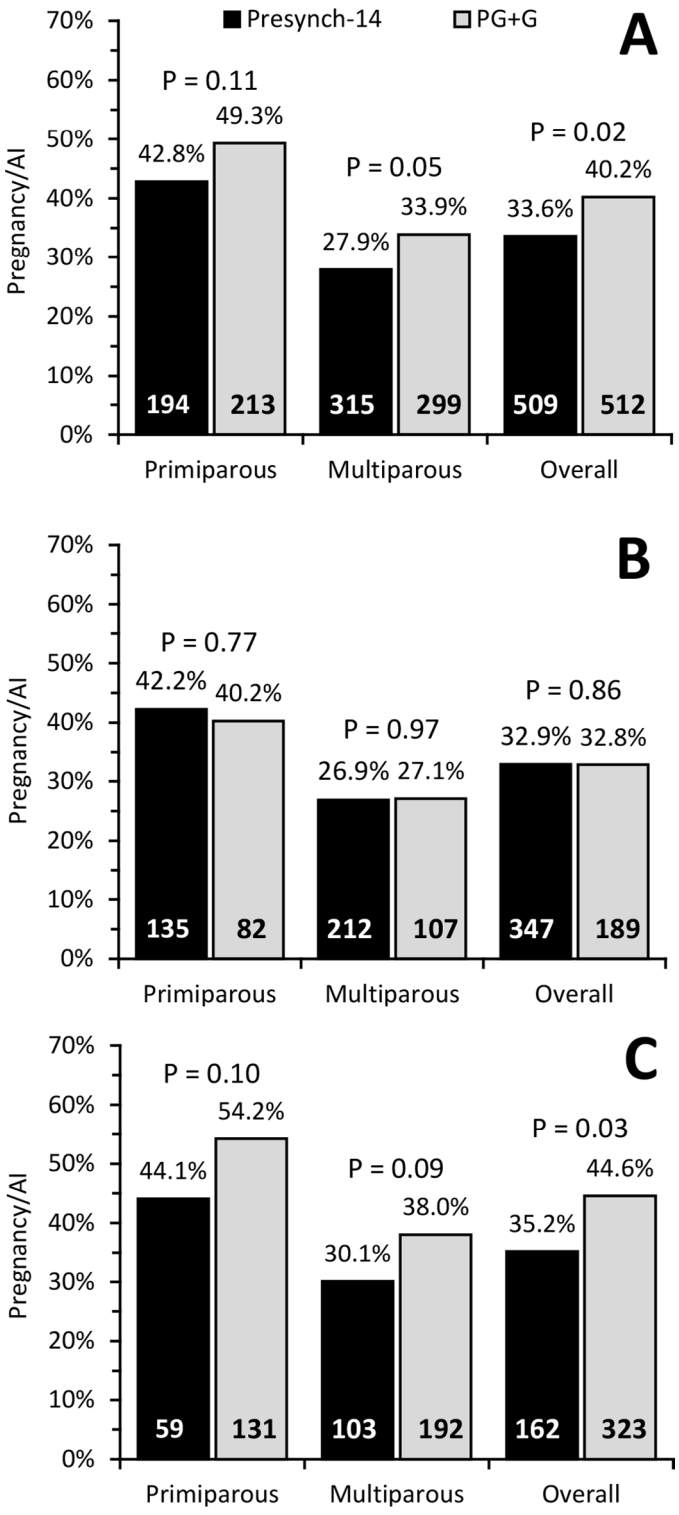


Figure 4. Effect of treatment, Presynch-14 (control) versus PG+G, on (A) pregnancy per AI on 42 ± 7 d post-AI for all cows, (B) cows only receiving AI following estrus detection, and (C) cows only receiving timed AI. Presynch-14 = a presynchronization strategy with 2 $PGF_{2\alpha}$ treatments 14 d apart with the last $PGF_{2\alpha}$ 14 d before the initiation of the Ovsynch protocol; PG+G = a simpler presynchronization program that uses $PGF_{2\alpha}$ and GnRH simultaneously 7 d before Ovsynch.

Two factors may justify the increased P/AI for cows treated with PG+G. First, PG+G doubled the proportion of cows receiving TAI compared with Presynch-14 (Figure 3), and cows receiving TAI had greater ($P < 0.01$) P/AI compared with cows receiving AIED (Table 4). A limitation of our study was that cows receiving AIED were inseminated earlier ($P < 0.001$), after the end of the VWP, compared with cows receiving TAI (64.0 ± 0.2 vs. 81.0 ± 0.2 DIM, respectively), which could have affected the P/AI outcome. Cows with earlier DIM at first service seem to have lower fertility compared with cows with later DIM at first service (Giordano et al., 2016). Nevertheless, Santos et al. (2017) demonstrated that cows at the same stage of lactation (~77 DIM) receiving TAI had greater P/AI for first service when compared with cows receiving AIED.

The second factor that may explain the effect of treatment on P/AI is the greater P/AI ($P = 0.03$) for PG+G than Presynch-14 in cows only receiving TAI (Figure 4). This increase seems to be due to the improvements in the ovarian physiological parameters before AI for cows receiving PG+G, including greater ovulatory response to G_1 and proportion of cows with functional CL and serum P4 at the $PGF_{2\alpha}$ of Ovsynch, and a lower proportion of cows with cysts at the $PGF_{2\alpha}$ of Ovsynch (Tables 1 and 2). The increase in the proportion of cows receiving AIED for Presynch-14 may have sorted cows with lower fertility to receive TAI. Nonetheless, we aimed to compare both strategies combined with AIED as a first-service reproductive breeding program.

Overall pregnancy loss from 42 ± 7 d post-AI to 245 d of gestation was 20.6% and did not differ among treatments or parities (Table 3). However, cows receiving AIED had a higher proportion of pregnancy losses than TAI (Table 4). The factors that may have affected P/AI differences between AIED and TAI may have also affected pregnancy loss differences between AIED and TAI. Chebel et al. (2004) did not find differences in pregnancy losses from 31 to 45 d after AI between AIED and TAI (overall losses = 12.5%). In our study, the first pregnancy diagnosis was performed later, between 35 and 49 d after AI. In addition, most pregnancy losses occurred between 71 to 140 d of gestation (d 0 of gestation = day of AI; Figure 5). We also speculated that cows receiving AIED might have a greater proportion of double ovulations and twin pregnancies, which may increase the proportion of pregnancy losses, especially in unilateral twin pregnancies (Martins et al., 2018; Garcia-Ispierto and López-Gatius, 2019; López-Gatius et al., 2020).

The proportion of cows calving per AI followed the same pattern as the P/AI results. Cows receiving PG+G had increased calving per AI compared

Table 4. Effect of AI type (AIED[1] vs. timed AI) on pregnancy per AI (P/AI), calving per AI, pregnancy loss, and twin births

	AI strategy		*P*-value		
Item	AIED[1]	Timed AI	AI strategy	Parity	AI strategy × parity
P/AI at 42 ± 7 d after AI, % (n/n)	32.8 (176/536)	41.4 (201/485)	<0.01	<0.001	0.99
Calving per AI,[2] % (n/n)	23.7 (125/527)	34.1 (164/481)	<0.001	<0.001	0.84
Pregnancy loss,[2] % (n/n)	25.2 (42/167)	16.8 (33/197)	0.04	0.11	0.49
Twin births, % (n/n)	16.0 (20/125)	14.6 (24/164)	0.76	0.72	<0.01

[1]AIED = artificial insemination after detection of estrus.

[2]Pregnancy losses from pregnancy diagnosis at 42 ± 7 d after AI to 245 d of gestation. Cows that left the herd after pregnancy diagnosis (n = 13) were not included in the analyses for calving per AI and pregnancy loss.

with Presynch-14. Primiparous cows also had greater ($P < 0.001$) calving per AI than multiparous cows (37.5 vs. 22.8%, respectively). Overall, cows receiving TAI also had a greater proportion of cows calving compared with AIED (Table 4). This difference resulted from improved P/AI at 42 d after AI and lower pregnancy losses for cows receiving TAI compared with AIED (Table 4).

The proportion of twin calvings did not differ between treatments or parities (Table 3), but it was higher (overall = 15.2%) than previously reported for lactating dairy cows in US herds [2.7 to 7.0% (Bicalho et al., 2007); 4.2% (del Río et al., 2007); and 6.4% (Schambow et al., 2021)]. This high proportion of twin calvings could be due to a combination of factors, including the high milk yield of cows used in the study (Lopez et al., 2005) and the stage of lactation and season (summer/fall in California) when cows were inseminated (Andreu-Vázquez et al., 2012; Schambow et al., 2021). An interaction parity × AI type (AIED vs. TAI) was observed for twin calvings (Table 4). Primiparous cows that received AIED had a greater ($P = 0.03$) proportion of twin births than TAI primiparous cows (20.9 vs. 8.3%, respectively). Conversely, multiparous cows that received AIED tended to have a lower ($P = 0.10$) proportion of twin births than multiparous cows that received TAI (10.3 vs. 21.3%, respectively). This result may be related to differences in circulating P4 concentrations and follicle dynamics between primiparous and multiparous cows before AIED and TAI.

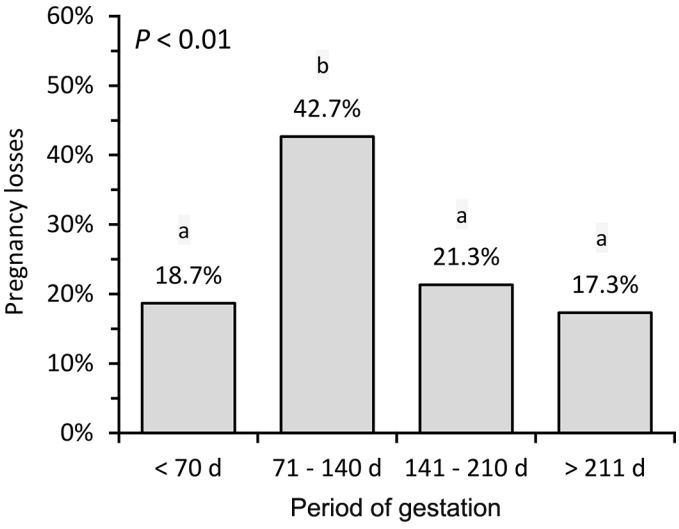


Figure 5. Distribution (% of total) of lactating dairy cows with pregnancy loss after pregnancy diagnosis at 42 ± 7 d across different periods of gestation. Columns with different letters (a, b) differ ($P < 0.05$).

CONCLUSIONS

The PG+G increased first-service P/AI and calving per AI compared with Presynch-14 when both treatments were used combined with AIED. The increase in P/AI and calving per AI for cows in PG+G appears to be due to the decrease in the proportion of cows receiving AIED and the improvements in the physiological parameters for cows receiving TAI. Cows receiving TAI were inseminated later in lactation, had greater P/AI and calving per AI, and had reduced pregnancy loss compared with cows receiving AIED.

ACKNOWLEDGMENTS

Thanks to the University of California Agriculture and Natural Resources, Cooperative Extension, Tulare County, and the University of Wisconsin–Madison for funding this project. A special thanks to Charles and Daniel Vanderkooi (Riverdale, CA) for allowing the use of their cows and facilities. We are also grateful to all the staff from Vanderkooi Dairy (Riverdale, CA) for their assistance during the data collection of this experiment.

We also thank Merck Animal Health (Madison, NJ) for donating Fertagyl and Estrumate used in this study. The authors have not stated any conflicts of interest.

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